

API driven e-store to sell vouchers & other digital products

Phase 1

- product - CRUD
- categories -CRUD
- Order -CRUD
- cancel order (complete order)
- refund (partial)
- Email for Order
- Email for invoice (WEB/PDF)

Phase 2

- Discount codes
- product bundles
- others

1. Users Table : Stores customer information.

```
CREATE TABLE Users (
    tenant_id int(10) UNSIGNED NOT NULL COMMENT 'Tenant ID',
    user_id BIGINT(15) NOT NULL,
    username VARCHAR(50) NOT NULL UNIQUE,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    image VARCHAR(256) NOT NULL UNIQUE,
    phone VARCHAR(100) NOT NULL UNIQUE,
    email VARCHAR(100) NOT NULL UNIQUE,

    user_type ENUM('INDIVIDUAL', 'COMPANY') NOT NULL DEFAULT 'INDIVIDUAL',
    company_name VARCHAR(100), -- Only applicable for COMPANY users
    company_registration_number VARCHAR(50), -- Only applicable for COMPANY
users
    company_address1 VARCHAR(255), -- Only applicable for COMPANY users
    company_address2 VARCHAR(255), -- Only applicable for COMPANY users
    company_country VARCHAR(255), -- Only applicable for COMPANY users
    company_pincode VARCHAR(255), -- Only applicable for COMPANY users
    tax_id VARCHAR(50), -- Only applicable for COMPANY users

    otp VARCHAR(8) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    status VARCHAR(2) NOT NULL, -- -1 INACTIVE, 0 ACTIVE
    created_by VARCHAR(2) NOT NULL,
    created TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_by VARCHAR(2) NOT NULL,
    updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    PRIMARY KEY (tenant_id,user_id)
);
```

```
CREATE TABLE Currencies (
    tenant_id int(10) UNSIGNED NOT NULL COMMENT 'Tenant ID',
    currency_code CHAR(3) NOT NULL, -- e.g., USD, EUR, SGD
    currency_name VARCHAR(50) NOT NULL,
    is_default VARCHAR(1) NOT NULL
    exchange_rate decimal(10,4) NOT NULL,
    symbol VARCHAR(10) NOT NULL,
    status VARCHAR(2) NOT NULL, -- -1 INACTIVE, 0 ACTIVE
    created_by VARCHAR(2) NOT NULL,
    created TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_by VARCHAR(2) NOT NULL,
    updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
```

```
    PRIMARY KEY (tenant_id,currency_code)
);
```

2. Categories Table

(Optional) Groups digital products into categories.

```
CREATE TABLE Categories (
    tenant_id int(10) UNSIGNED NOT NULL COMMENT 'Tenant ID',
    category_id BIGINT(15) NOT NULL,
    category_name VARCHAR(50) NOT NULL,
    description TEXT,
    status VARCHAR(2) NOT NULL, -- -1 INACTIVE, 0 ACTIVE
    created_by VARCHAR(2) NOT NULL,
    created TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_by VARCHAR(2) NOT NULL,
    updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    PRIMARY KEY (tenant_id,category_id)
);
```

```
CREATE TABLE CategoryTranslations (
    tenant_id int(10) UNSIGNED NOT NULL COMMENT 'Tenant ID',
    category_id BIGINT(15) NOT NULL,
    language_code CHAR(2) NOT NULL,          -- e.g., 'en', 'fr', 'zh'
    category_name VARCHAR(50) NOT NULL,
    description TEXT,
    status VARCHAR(2) NOT NULL, -- -1 INACTIVE, 0 ACTIVE
    created_by VARCHAR(2) NOT NULL,
    created TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_by VARCHAR(2) NOT NULL,
    updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    PRIMARY KEY (tenant_id,category_id,language_code)
    --PRIMARY KEY (category_id, language_code),
    --FOREIGN KEY (category_id) REFERENCES Categories(category_id)
);
```

3. Products Table

General product information for digital items.

```
CREATE TABLE Products (
    tenant_id int(10) UNSIGNED NOT NULL COMMENT 'Tenant ID',
    product_id INT BIGINT(15) NOT NULL,
    product_name VARCHAR(100) NOT NULL,
    description TEXT,
    default_price DECIMAL(10, 2) NOT NULL, -- Price in default currency (for
reference only)
    default_currency CHAR(3) NOT NULL,      -- Default currency code (should
exist in Currencies)
    category_id INT,
    status VARCHAR(2) NOT NULL, -- -1 INACTIVE, 0 ACTIVE
    created_by VARCHAR(2) NOT NULL,
    created TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_by VARCHAR(2) NOT NULL,
    updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    PRIMARY KEY (tenant_id,product_id)
    --FOREIGN KEY (category_id) REFERENCES Categories(category_id)
);
```

```
CREATE TABLE ProductTranslations (
    tenant_id int(10) UNSIGNED NOT NULL COMMENT 'Tenant ID',
```

```

product_id BIGINT(15) NOT NULL,
language_code CHAR(2) NOT NULL,          -- e.g., 'en', 'fr', 'zh'
product_name VARCHAR(100) NOT NULL,
description TEXT,
status VARCHAR(2) NOT NULL, -- -1 INACTIVE, 0 ACTIVE
created_by VARCHAR(2) NOT NULL,
created TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_by VARCHAR(2) NOT NULL,
updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
PRIMARY KEY (tenant_id, product_id, language_code)

--FOREIGN KEY (product_id) REFERENCES Products(product_id)
);

```

4. DigitalProductDetails Table

Contains details specific to digital products, such as file URLs and license info.

```

CREATE TABLE DigitalProductDetails (
    tenant_id int(10) UNSIGNED NOT NULL COMMENT 'Tenant ID',
    product_id BIGINT(15) NOT NULL,
    file_url VARCHAR(255) NOT NULL, -- could be an S3 URL or similar
    file_size INT, -- size in KB/MB
    file_format VARCHAR(20), -- e.g., PDF, MP3, MP4, etc.
    license_info TEXT, -- terms or license keys
    version VARCHAR(20),
    status VARCHAR(2) NOT NULL, -- -1 INACTIVE, 0 ACTIVE
    created_by VARCHAR(2) NOT NULL,
    created TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_by VARCHAR(2) NOT NULL,
    updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    PRIMARY KEY (tenant_id, product_id)
    --FOREIGN KEY (product_id) REFERENCES Products(product_id)
);

```

5. Orders Table

Records a customer's order.

```

CREATE TABLE Orders (
    tenant_id int(10) UNSIGNED NOT NULL COMMENT 'Tenant ID',
    order_id INT BIGINT(15) NOT NULL,
    user_id INT NOT NULL,
    order_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    currency VARCHAR(3),
    total_amount DECIMAL(10, 2) NOT NULL,
    exchange_rate DECIMAL(10, 2) NOT NULL,
    status VARCHAR(20) DEFAULT 'Pending', -- e.g., Pending, Completed, Cancelled

    created_by VARCHAR(2) NOT NULL,
    created TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_by VARCHAR(2) NOT NULL,
    updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    PRIMARY KEY (tenant_id, order_id)
    --FOREIGN KEY (user_id) REFERENCES Users(user_id)
);

```

6. OrderItems Table

Lists the individual products purchased within an order.

```

CREATE TABLE OrderItems (
    tenant_id int(10) UNSIGNED NOT NULL COMMENT 'Tenant ID',
    order_id INT NOT NULL,
    order_item_id BIGINT(15) NOT NULL,

    product_id INT NOT NULL,
    price_at_purchase DECIMAL(10, 2) NOT NULL,
    license_key VARCHAR(100), -- if a unique key is generated per order
    status VARCHAR(2) NOT NULL, -- -1 INACTIVE, 0 ACTIVE
    created_by VARCHAR(2) NOT NULL,
    created TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_by VARCHAR(2) NOT NULL,
    updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,

    PRIMARY KEY (tenant_id, order_id, order_item_id)
    --FOREIGN KEY (order_id) REFERENCES Orders(order_id),
    --FOREIGN KEY (product_id) REFERENCES Products(product_id)
);

```

7. Payments Table

Captures payment information associated with orders.

```

CREATE TABLE Payments (
    tenant_id int(10) UNSIGNED NOT NULL COMMENT 'Tenant ID',
    payment_id INT BIGINT(15) NOT NULL,
    order_id INT NOT NULL,
    currency VARCHAR(3),
    payment_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    amount DECIMAL(10, 2) NOT NULL,
    payment_method VARCHAR(50), -- e.g., Credit Card, PayPal
    transaction_id VARCHAR(100),
    status VARCHAR(20) DEFAULT 'Successful', failed

    created_by VARCHAR(2) NOT NULL,
    created TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_by VARCHAR(2) NOT NULL,
    updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    PRIMARY KEY (tenant_id, payment_id)
    --FOREIGN KEY (order_id) REFERENCES Orders(order_id)
);

```

8. DigitalDownloads Table

Tracks each time a digital product is downloaded by a customer.

```

CREATE TABLE DigitalDownloads (
    tenant_id int(10) UNSIGNED NOT NULL COMMENT 'Tenant ID',
    download_id BIGINT(15) NOT NULL,
    order_item_id INT NOT NULL,
    download_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ip_address VARCHAR(45),
    status VARCHAR(20) DEFAULT '0', --0:ACTIVE , -1 INACTIVE

    created_by VARCHAR(2) NOT NULL,
    created TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_by VARCHAR(2) NOT NULL,
    updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    PRIMARY KEY (tenant_id, download_id)
    --FOREIGN KEY (order_item_id) REFERENCES OrderItems(order_item_id)
);

```

Optional: Reviews Table

Allows customers to review products.

```
CREATE TABLE Reviews (  
    tenant_id int(10) UNSIGNED NOT NULL COMMENT 'Tenant ID',  
    review_id BIGINT(15) NOT NULL,  
    product_id INT NOT NULL,  
    user_id INT NOT NULL,  
    rating INT CHECK (rating BETWEEN 1 AND 5),  
    comment TEXT,  
    status VARCHAR(20) DEFAULT '0', --0:ACTIVE , -1 INACTIVE  
  
    created_by VARCHAR(2) NOT NULL,  
    created TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    updated_by VARCHAR(2) NOT NULL,  
    updated TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  
    PRIMARY KEY (tenant_id,review_id)  
    --FOREIGN KEY (product_id) REFERENCES Products(product_id),  
    --FOREIGN KEY (user_id) REFERENCES Users(user_id)  
);
```

Explanation of Updates

Currencies Table:

A new table that holds supported currencies. Both the Products table (via default_currency) and Orders table (via currency_code) reference this table.

Multi- Currency Pricing:

The ProductPrices table allows you to define the price of each product in one or more currencies. This table is linked to Products and Currencies.

Multi- Language Support:

The ProductTranslations and CategoryTranslations tables store locale- specific product and category names/descriptions. Each record includes a language_code so that the system can present the correct text based on the user's locale.

Orders & Payments:

The Orders table now stores a currency_code to capture the currency used for that order. The Payments table remains largely unchanged, assuming payments align with the order currency.

This updated schema enables an e- store to offer digital products with multi- currency pricing and multi- language descriptions, making it adaptable for a global audience while supporting both individual (B2C) and business (B2B) users.

Resources

<https://www.youtube.com/watch?v=yp28g5AJczM>

Swagger to generate API Documentation

<https://www.baeldung.com/swagger-2-documentation-for-spring-rest-api>

spring jpa

hibernate

mysql

Spring security (OAuth 2)

Stripe : <https://www.youtube.com/watch?v=BczS-wbxgp4>